

Nothing is allowed to spend money before something checks it can

A spend ceiling that runs before the first vendor call, a ledger that records what was reserved and refused, and a reader in the control room that says what a check actually costs and what that figure is built on.

01 What was wrong

The route that spends money had a rate limit of five requests a minute, held in a map in memory, per instance, reset on every cold start. Its own comment said so: **move to Vercel KV before this URL is public**. It counted requests. It did not count dollars, and nothing anywhere counted dollars.

Two consequences followed from that, and neither was visible from the board.

A request cost whatever it cost	Five requests a minute is a ceiling on volume. With a search clamp of 34 and a reasoning call of 16,000 output tokens, five requests a minute is not a ceiling on spend, and the two were being treated as the same number.
The documented ceiling was wrong	<code>.env.example</code> said the search clamp was 3 to 14 with a default of 10. The code clamped 3 to 34 and defaulted to 22. Anyone sizing the bill off the file was out by more than half.

02 What was built

<code>api/_budget.js</code>	One module. <code>reserve()</code> decides before the first vendor call and records the reservation as it decides, so a burst of concurrent requests cannot all read the same low count and pass. <code>settle()</code> writes what the run actually cost against what was reserved.
<code>db/spend.sql</code>	The ledger. Salted client key, timestamp, estimate, measurement. No address, no identifier, no party. Eight day purge. Not yet run against the database.
<code>api/admin-metrics.js</code>	A spend block on the existing route. No second door, no second auth path.
<code>admin.html</code>	A spend pill on the reader row, and a reader behind it.
<code>tools/budgetcheck.mjs</code> <code>tools/spendcheck.mjs</code>	Twenty-nine assertions between them. Both in <code>npm test</code> .

Three counters, not one

Runs per window, dollars per window, dollars per day, all per client, plus a house ceiling for the day across everyone. Any one of them refuses.

The house ceiling never escalates. A reader who arrives on the day it is reached has done nothing wrong, and treating them as though they had is how a product punishes somebody for its own budget.

Volume throttles. Repeat behaviour blocks.

A client refused three times inside fifteen minutes is paused, and the pause doubles each time it happens again, to a six hour cap. One oversized request is refused and never paused, because size is not behaviour. There is a test that fails if that inverts.

It never falls open. Where Postgres is unconfigured or fails, the in-memory ledger carries the same rules on a single instance and the refusal still fires. Storage must never be able to break a check, and must never be able to let one through either.

api/_budget.js, header comment

03 What a check costs

A typical party runs 18 Exa searches, 11 of them asking for full text, and 4 Parallel objectives. At the published list rates that is **\$0.212**.

COMPONENT	RATE	PER CHECK	STATUS
Exa search	\$7 per 1,000	\$0.126	BILLED
Exa full text	\$1 per 1,000 pages	\$0.066	BILLED
Parallel	\$5 per 1,000	\$0.020	LIST RATE
Claude	per million tokens	not measured	DEFAULT RATE

Exa returns its own billed figure on every call and that figure is stored, so retrieval is an invoice number rather than a rate applied to a count. The reasoning half is not. No live run has had its token counts read back, and the rates in force are defaults I set rather than figures read off the invoice.

Until KBYS_RATE_CLAUDE_IN_M and KBYS_RATE_CLAUDE_OUT_M are set on the deployment, every dollar in the control room carries a line saying it is priced from a default. The screen is not allowed to let an estimate wear the clothes of a measurement.

The first band of the spend reader, above every figure on it

The ceiling in force

CEILING	VALUE	WHAT IT MEANS
---------	-------	---------------

Per minute	5 checks, \$3.75	Whichever bites first. At \$0.74 reserved per check the two land together.
Per client, per day	\$12.00	About 16 checks.
House, per day	\$120.00	Checking pauses for everyone until midnight.
Block	3 refusals, 15 minutes	15 minute pause, doubling, 6 hour cap.

The reservation is the ceiling of a check, not its average: 28 searches, 4 objectives, 90,000 tokens in and 16,000 out, which is **\$0.74**. The average is what has to be affordable in aggregate; the ceiling is what has to be affordable when one request goes long.

04 The spend reader

A third pill sits on the operations row beside New and the fix list, carrying today's spend. It is never red for spending money. It warms as the day's house ceiling fills, and only the last band raises its voice, because that is the only band where anything is about to stop.

BAND	WHAT IT CARRIES
The rate card	What the dollars below are priced from. First, because everything under it depends on it being true.
Today	Spent, left before checking stops, the house ceiling, tokens in and out, and a bar for the day.
What one check costs	Average, median, ninth in ten, dearest. Tokens in and out per check. Reasoning against retrieval as a split. An average on its own hides the run that cost four times the rest.
The last 30 days	Total, the two halves, average wait, and the last ten days one row each.
The ceiling	Every limit in force, how many checks were refused today and why, and how many clients are paused now.

Where a run recorded no billed figure from Exa it is counted at nought for retrieval and the count of such runs is printed, rather than blending an estimate into a measurement. Where no run has been recorded at all, the reader says there is nothing to average and refuses to produce one.

05 How it was proved

Every fix has a test that fails if the fix is reverted. Each was reverted and the failure observed.

REVERTED	WHAT FAILED
The refusal on a failed reservation	budgetcheck, 2 lines

The gate moved to after retrieval	budgetcheck, all 4 vendor calls
The spend pill removed	spendcheck, 1 line
The rate card warning removed	spendcheck, 1 line
The spend block no longer served	spendcheck, 1 line

The ordering test is guarded against passing vacuously: if it can no longer find at least three vendor calls in the route to order against, it fails rather than reporting success on an empty list.

06 **What is not done**

The Claude rate card	OPEN Two figures off the Anthropic console, set as environment variables on the deployment. Until then the dollar ceilings are approximately right, not exactly right.
db/spend.sql	OPEN Written, not run. The ceiling enforces without it, per instance.
db/telemetry.sql migration	OPEN The not_asked and computed columns are still waiting on the same run.
Steps 3 to 8	OPEN Response shaping, untrusted content delimiting, scheduled route auth, SQL and error handling, the client bundle scan, and the compliance gate with CI.